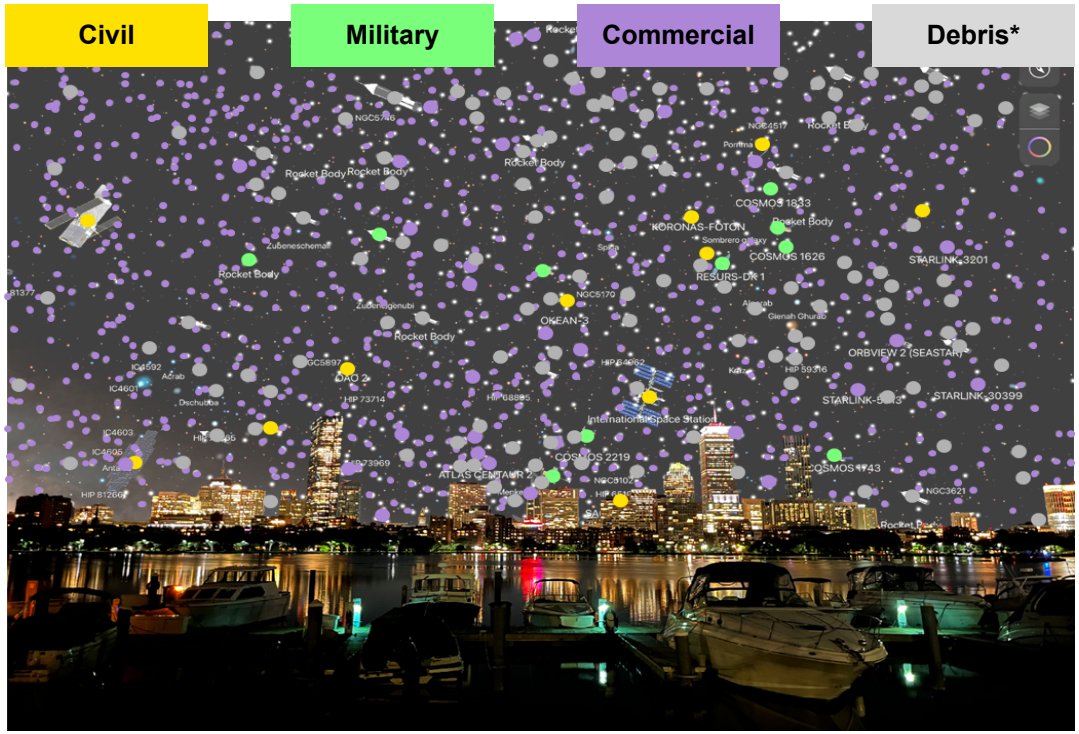
Strategic Sensor Tasking for SDA using Graph Neural Networks

Rhea Senthil Kumar

June 2025

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MASSACHUSETTS INSTITUTE OF TECHNOLOGY

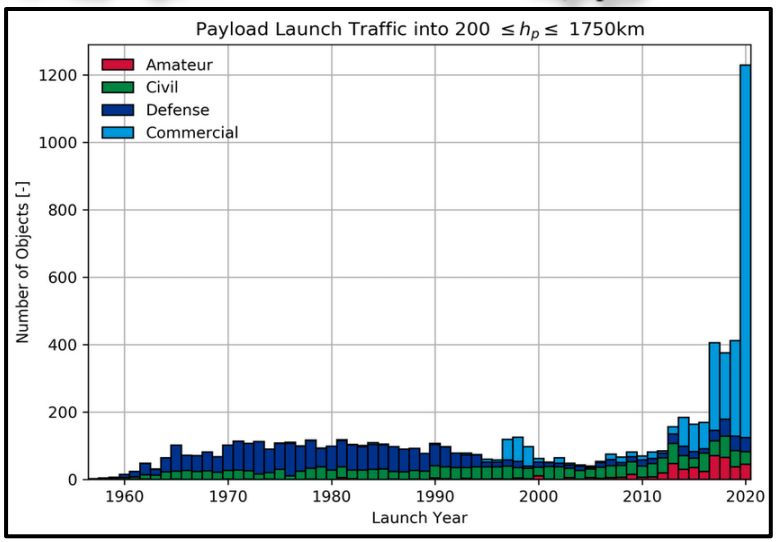
Congestion of Low Earth Orbit



Boston, MA

Source: Night Sky (app) 2024.

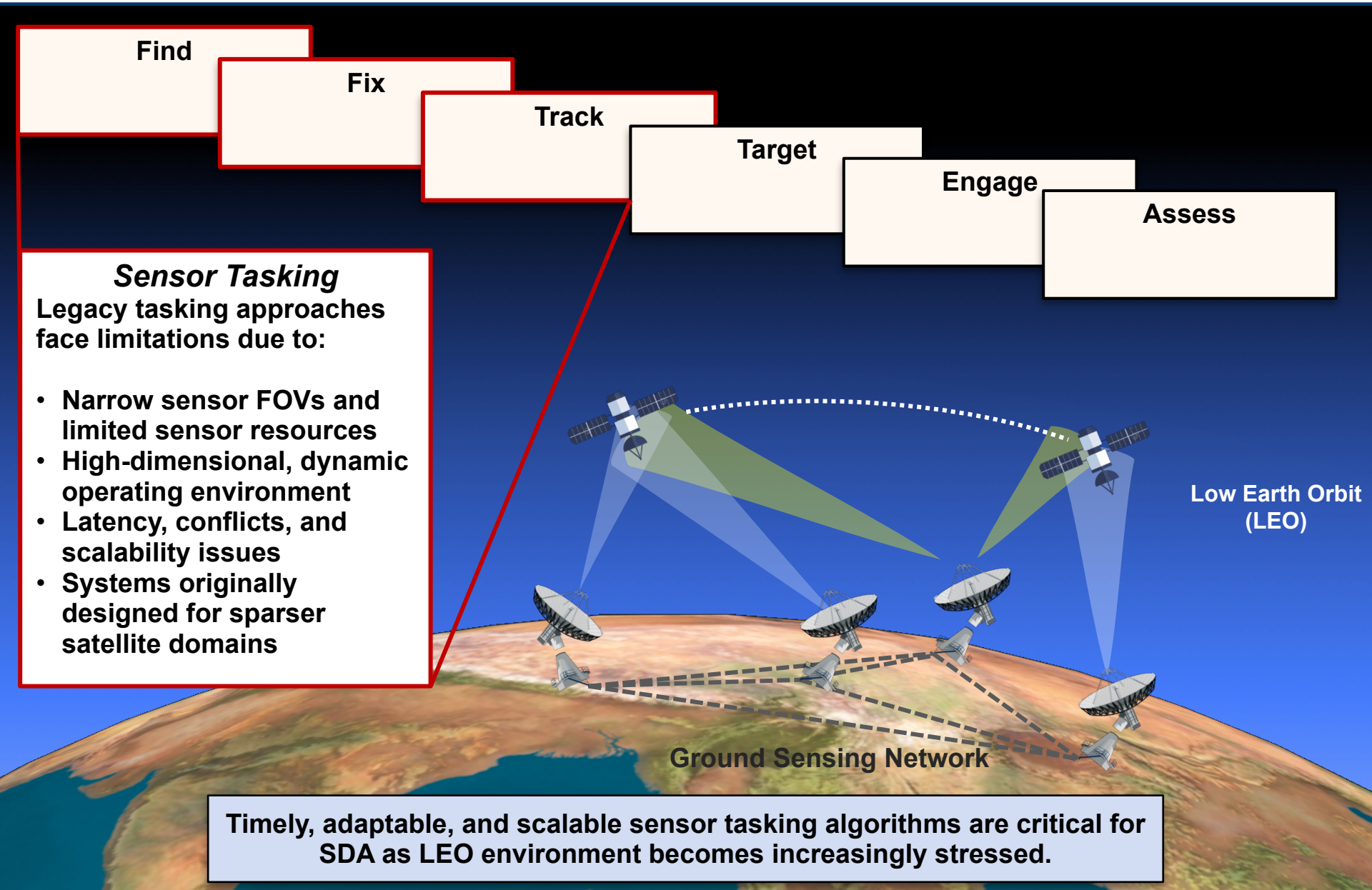
> "… the space sector is growing and evolving at a very rapid pace … more human made objects, satellites, space systems and installations will be in the Earth orbits, causing its congestion, especially in Low Earth Orbit (LEO) ."
>
> Defense Science Review (2022).



- **Commercialization has led to rapid growth in LEO objects**
- **Space Domain Awareness (SDA) is increasingly challenging, and complexity grows as more satellites are launched into LEO**

As LEO becomes more congested, efficient space sensing capabilities are necessary to maintain custody of space assets.

Sensor Tasking for SDA



Approaches to Sensor Tasking



Brute Force

→ Systematically follow fixed sweeping patterns

Pro:
- Quick baseline
- Complete search

Cons:
- No adaptability to dynamics
- No scalability to larger systems

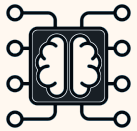


Heuristics

→ Simple algorithmic rules or use last known target position

Pro:
- Use a priori information to choose actions

Cons:
- Poor scalability to large systems
- Brittle to system changes



Machine Learning (ML) Frameworks

→ Learn optimal policies by simulating interactions

Pros:
- Learns and responds to changing environments
- Scalable policies

Cons:
- Requires training infrastructure
- High initial compute cost

Machine learning has capability to anticipate target behavior and learn to respond dynamically as opposed to traditional search methods.

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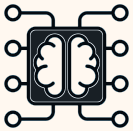


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Outline

- Introduction
- Simulation Framework Overview
- Performance Metrics
- Summary and Path Forward

Scenario Overview

CONUS Sensor Network



→ **Explore the feasibility and value of learning-based tasking policies for CONUS SSN sensors**

- **Back-of-the-envelope LOS* analysis**
- **Study's focus is relative algorithm performance and behavioral trends**

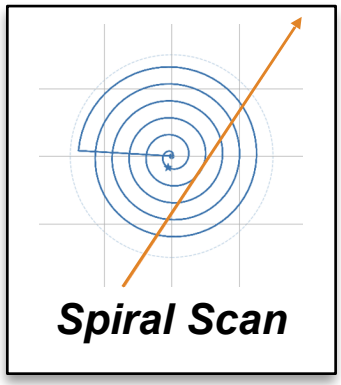
What This Analysis Does Not Include:

- **Sensor-specific hardware modelling**
- **Orbital perturbations, atmospheric effects**
- **Communication latency**
- **Earth geometry or occlusion**
- **Policy retraining or online learning**

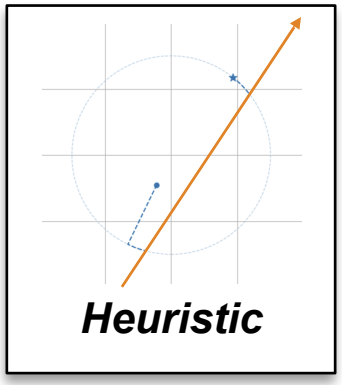
Study assesses ML's ability to learn cooperative, adaptive tasking strategies, which is ideal for dynamic, congested LEO environments.

**Line-of-Sight*

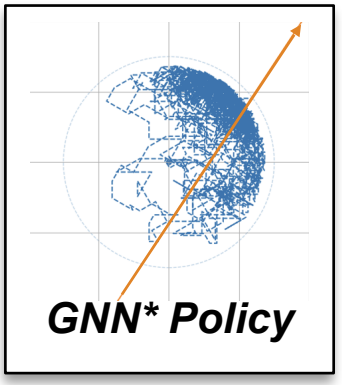
Algorithms and Metrics of Interest



- Brute force spiral search
- No prior target information



- Prior expected target track direction
- Greedy scheduling



- Expected target track in feature set
- Dynamic scheduling over multiple episodes

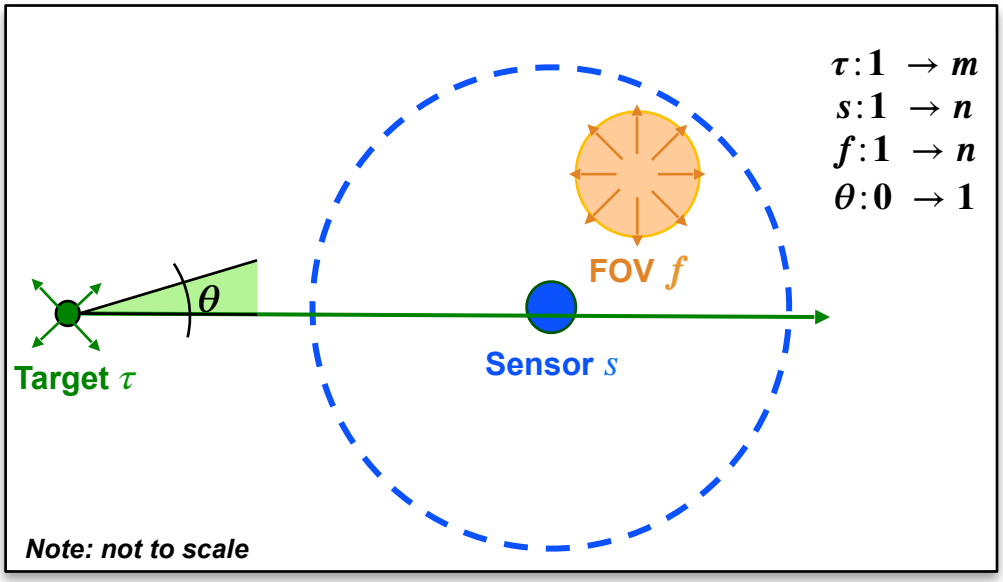
Increasing coordination

Core Objectives of SDA and Relevant Metrics:

- Persistent custody of full object catalog
  - Proportion of targets detected
- Timely initial detection is critical to catch targets during pass (~10 min)
  - Time to first detection
- High-precision state estimation requires sustained tracking
  - System-level detection rate

***Graph Neural Network**

Simulation Setup



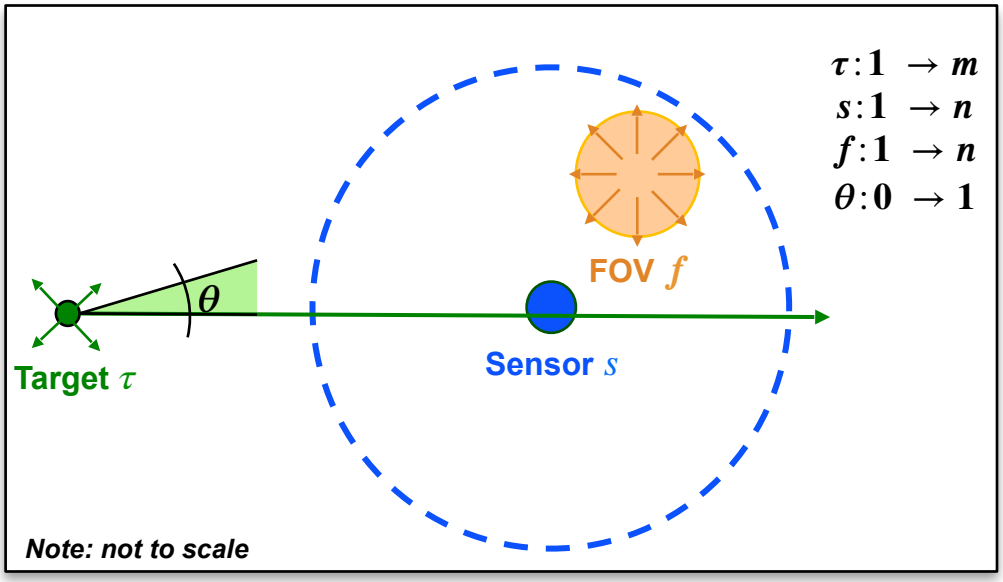
Parameter Inputs

Target Maneuvers	• Up to ±1° cross-track maneuvers during orbit
Target Density	• Undersaturated, saturated, oversaturated
Sensor Model	• 10° elevation constraint • 1° FOV • 5 deg/s max slew rate

GNN Policy Notes:

- Sensors coordinate via neighboring edges and central global node
- Node feature set includes position, velocity, slew rate, mode flag (search vs. track)

Simulation Setup



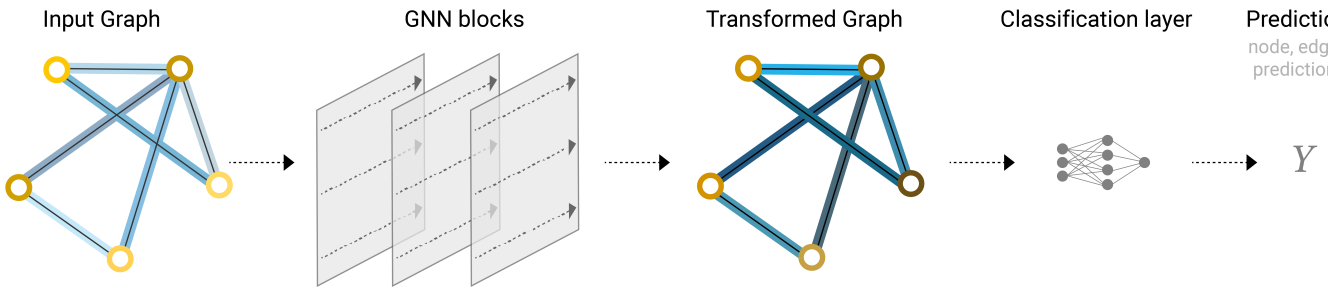
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- **GNN policy trained with reinforcement learning to optimize adaptive tasking behavior**



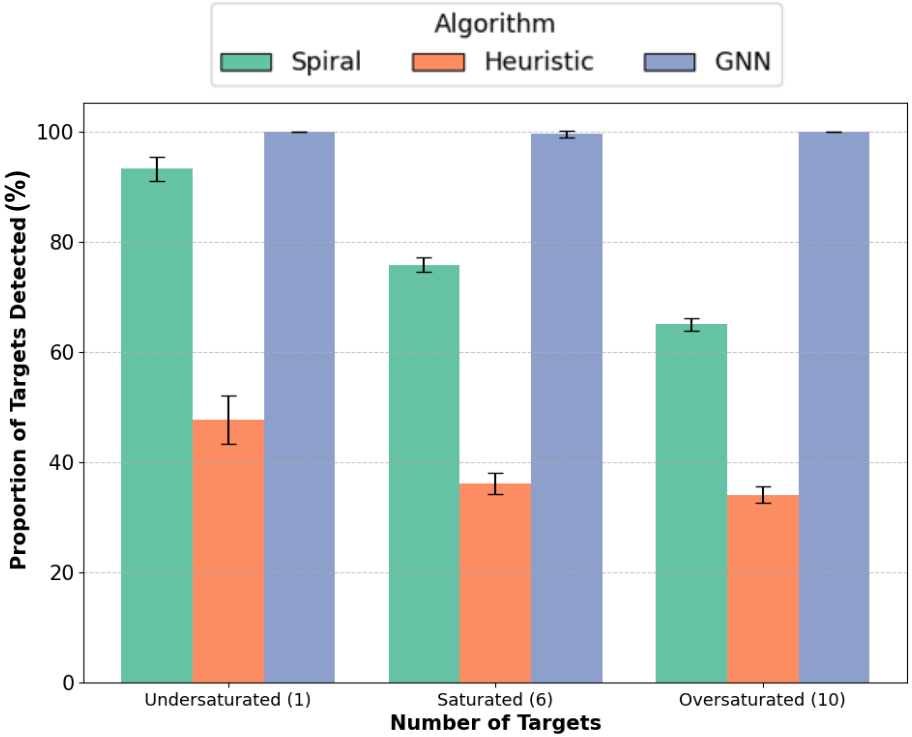
- **Representation of multi-agent systems**
- **Scalable reasoning over system structure and dynamics**

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Proportion of Targets Detected

- 500 Monte Carlo runs • 30 min episodes • 1 deg maneuver magnitude

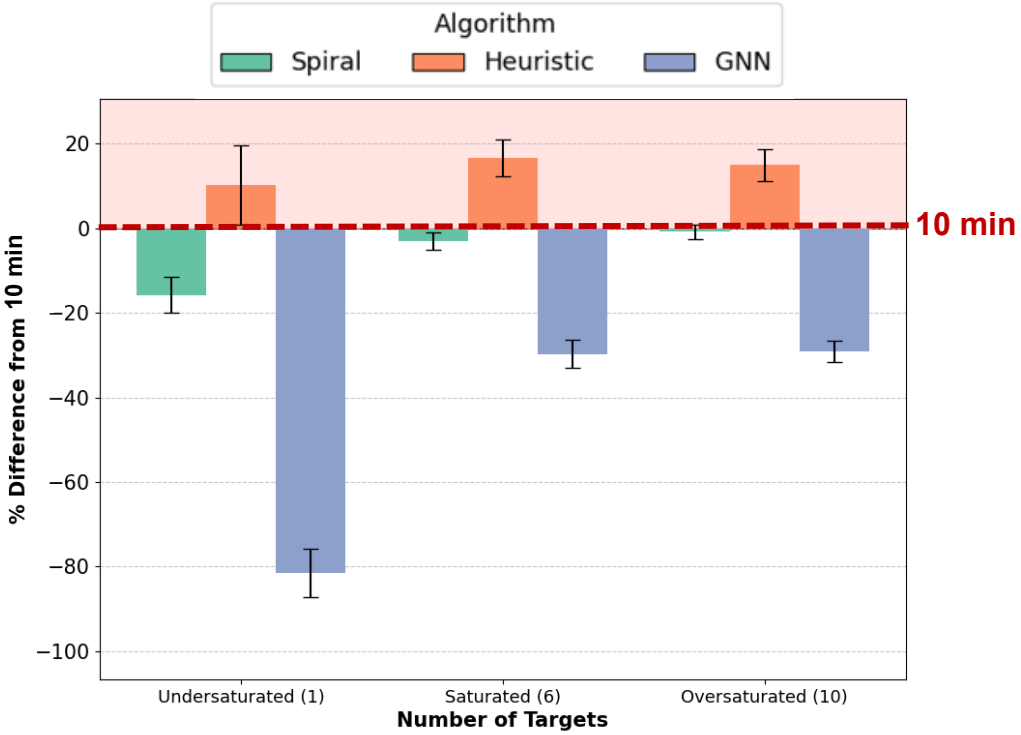


- Baseline algorithms degrade in oversaturated conditions
- Heuristic algorithm is highly sensitive to maneuvering and system load
- GNN achieves nearly perfect detection across all target densities

GNN-based approach outperforms traditional methods in presence of maneuvers, indicating strong generalization to dynamic scenarios and uncertainty.

Time to First Detection

- 500 Monte Carlo runs • 30 min episodes • 1 deg maneuver magnitude

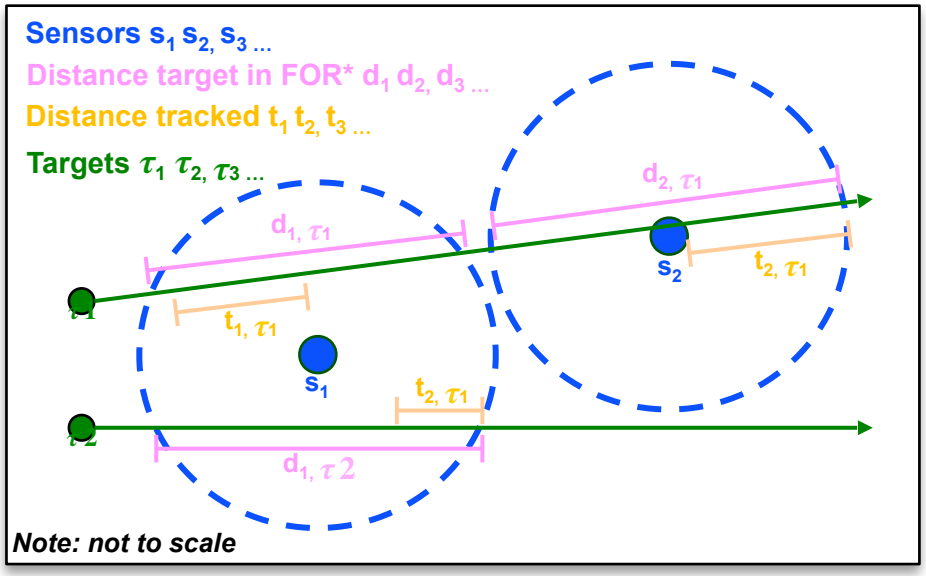


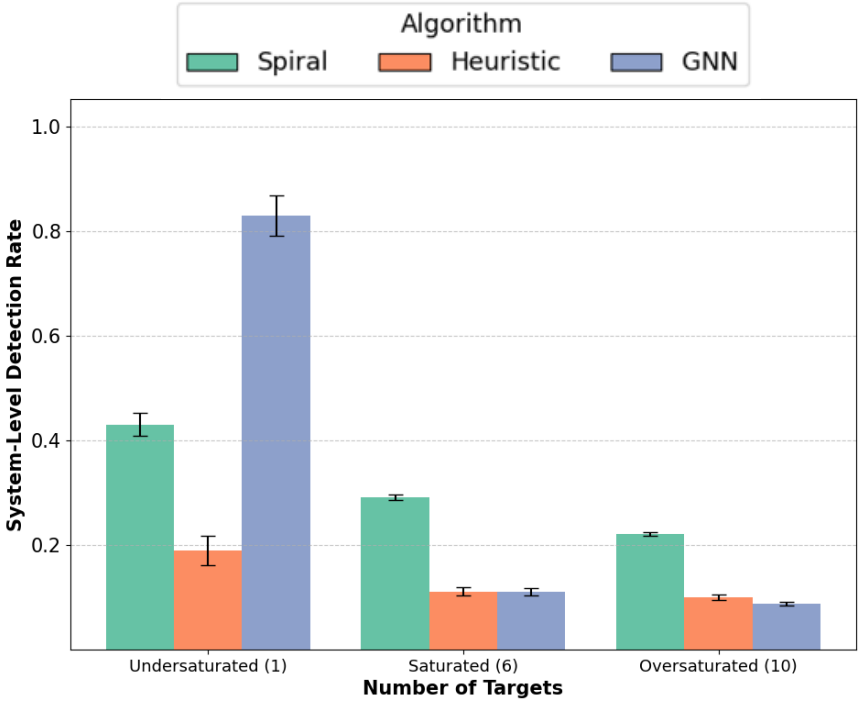
- Heuristic algorithm collapses under maneuvers, indicating missed passes and poor adaptability
- GNN achieves fastest detection across all target densities

GNN policy displays the earliest and most reliable detection for a 10-minute pass window.

System-Level Detection Rate

- 500 Monte Carlo runs • 30 min episodes • 1 deg maneuver magnitude





- Baselines achieve higher system-level detection by persistently tracking fewer targets
- GNN policy sacrifices detection consistency to prioritize catalog coverage

GNN policy learns when to switch targets strategically, maintaining comparable system-level detection rate to heuristic methods with higher catalog object visit rate.

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Performance Metric Overview

Fails SDA mission objectives | Further development required | Meets SDA mission objectives

Performance Metrics	Spiral Scan	Greedy Heuristic Algorithm	GNN Policy
Percentage of Catalog Tracked			
Time to First Detection			
System-Level Detection Rate			
Scalability to Target Density			
Runtime Efficiency			

- The GNN policy consistently outperforms baselines in catalog custody and detection speed, even under target maneuvering and increasing system complexity.
- Tradeoffs with baselines in system level detection rate and runtime efficiency

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Summary

- SDA is increasingly challenged by the rapid growth of space objects relative to available sensors, which strains traditional sensor tasking methods.
- GNNs offer a scalable, adaptive approach to persistent catalog custody in congested LEO environments.
  - Up to 70% higher catalog coverage than heuristic algorithm under high-density, maneuvering target scenarios
- Comparable performance is achieved even with simplified assumptions, demonstrating early promise for real-world deployment
- Further investment in autonomous and intelligent tasking software should be considered to advance resilient space domain awareness capability

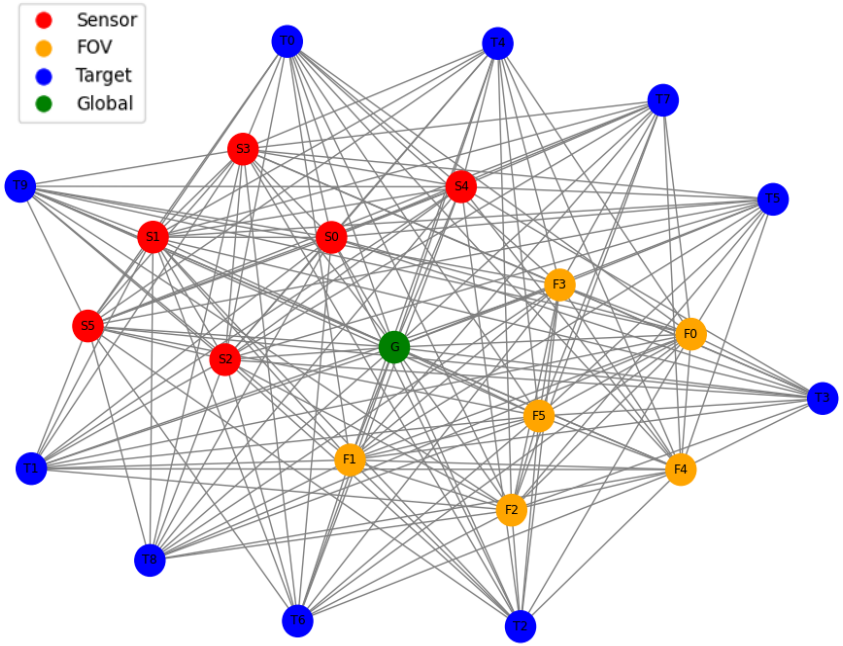
Path Forward

- **Integrate higher-fidelity sensing and orbital dynamics models, and evaluate algorithms at constellation scale (e.g. Starlink)**
- **Conduct sensitivity analyses on maneuver magnitude and frequency for mission-specific scenarios**
- **Explore tradeoffs between online and offline learning, alternative machine learning architectures, and reward tuning to optimize performance**
- **Benchmark against current tasking algorithms and pursue deployment of learning-based ML frameworks to real or near-operational systems**

Backup Slides



Model Visualization



Node Types:

- Sensor, FOV, Target, Global

Node Features:

- Longitude, latitude, slew rate, target speed, velocity components, mode flag (search v. track)
- Global node features are graph-level summary stats

Edges:

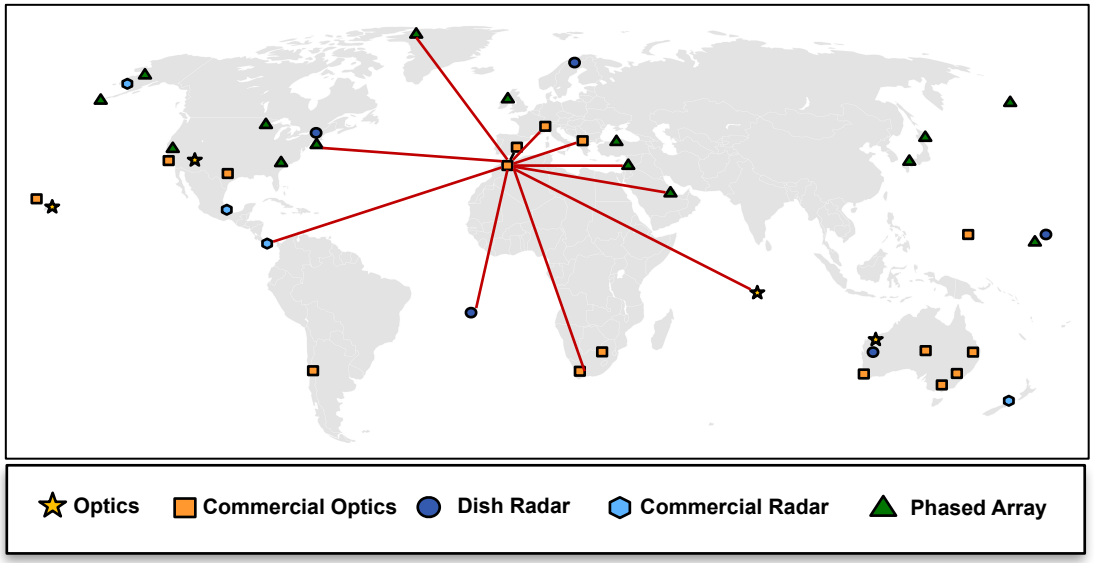
- Sensor ↔ Sensor
- Sensor ↔ FOV
- Sensor ↔ Targets
- FOV ↔ Targets
- FOV ↔ FOV
- Global node ↔ All other nodes

Summary:

- Sensors and FOVs model sensor positions and orientations
- Targets modeled with relative spatial features to sensors to allow message passing with spatial context
- Global node aggregates mission-level info, accessible to all nodes during message passing

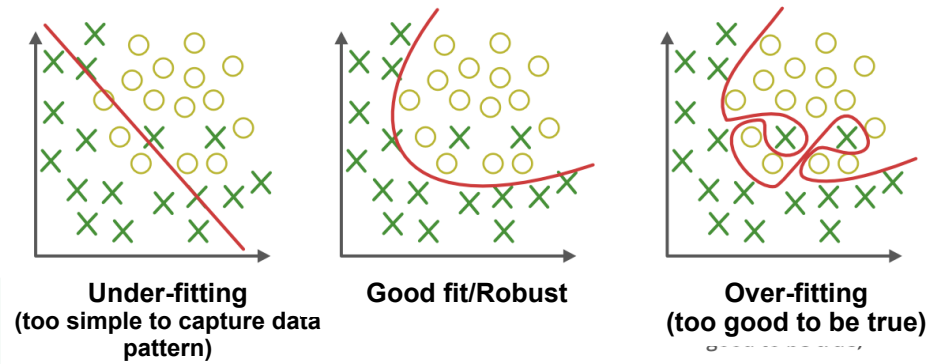
GNNs for Sensor Tasking

Ground-Based Space Surveillance Network (SSN)



Graph-based ML is naturally suited for sensor tasking:

- Natural representation of multi-sensor, multi-target system
- Scalability to system-level complexity
- Joint reasoning over structure and state
- Decentralized cooperation between sensors



Under-fitting (too simple to capture data pattern)

Good fit/Robust

Over-fitting (too good to be true)

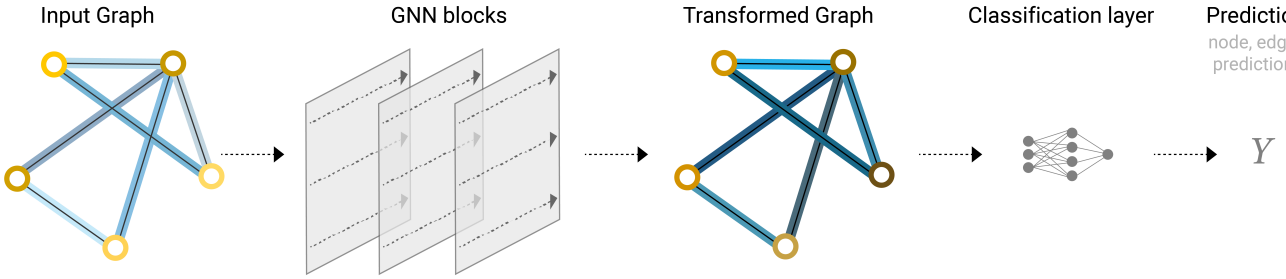
Limitations of ML-Only Approach:

- Tends to overfit to dominant strategies in training set; frequent iteration needed as LEO environment changes
- No incentive modeling; models cannot react to behavior to reach goals or abide by constraints

GNNs excel at learning complex object relationships, but lack inherent framework to reason over objectives or adapt to changing distributions.

Game-Theoretic Reinforcement Learning

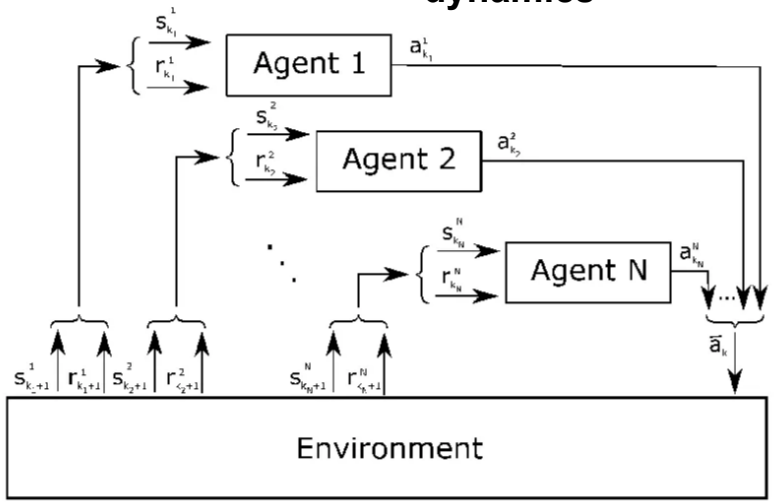
- **GNN policy trained with reinforcement learning to optimize adaptive tasking behavior**



- **Representation of multi-sensor, multi-target systems**
- **Decentralized cooperation across nodes**
- **Scalable reasoning over system structure and dynamics**

Reinforcement Learning:

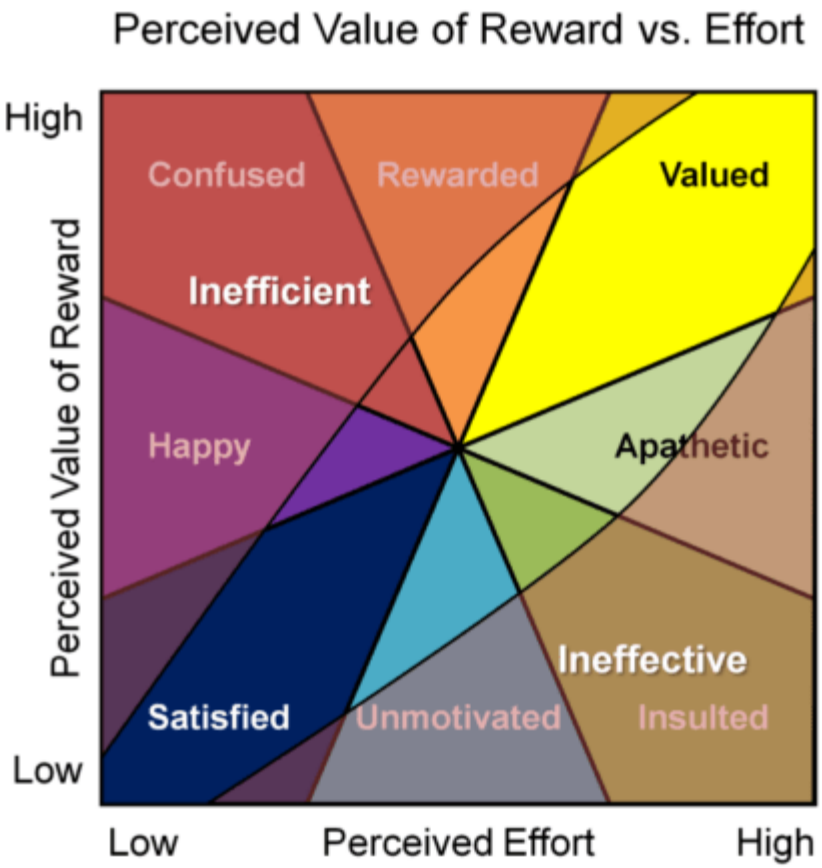
- **Well-suited for sequential decision-making problems where optimal strategy must be learned to achieve shared objective**
- **Decentralized, independent cooperation between multiple agents**
- **Equilibrium-oriented decision making under uncertainty**
  - **Exploration vs. Exploitation**



GTRL enables sensors to learn cooperative, adaptive tasking strategies, which is ideal for dynamic, congested LEO environments.

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GNN Reward Function



Function Components

Rewards:

- Switching from search to track mode (new detection
- Continuous steps in track mode
- Target is inside sensor FOR
- Coverage bonus when sensor direction differs from others

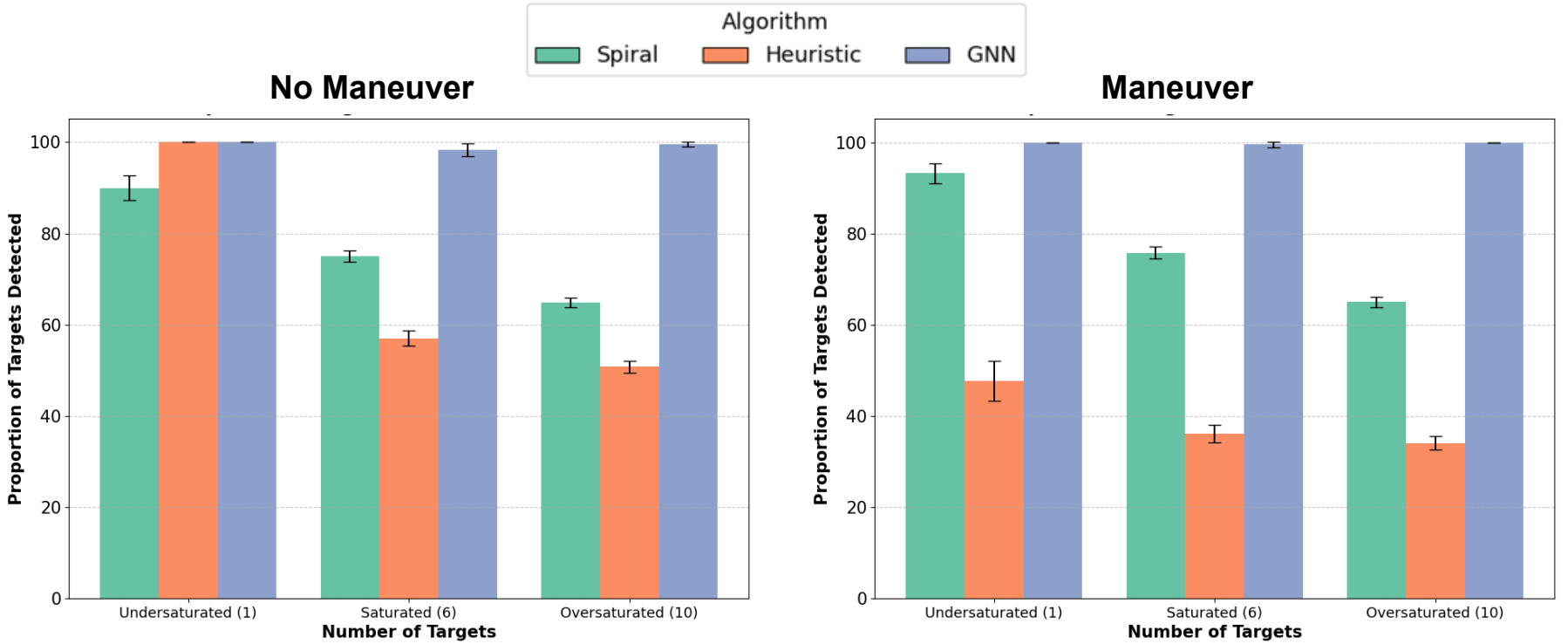
Penalties:

- Haversine distance to nearest target when in search mode
- Continuous steps in search mode

Constructed reward encourages fast target acquisition, persistent tracking, and coordinated sensor coverage.

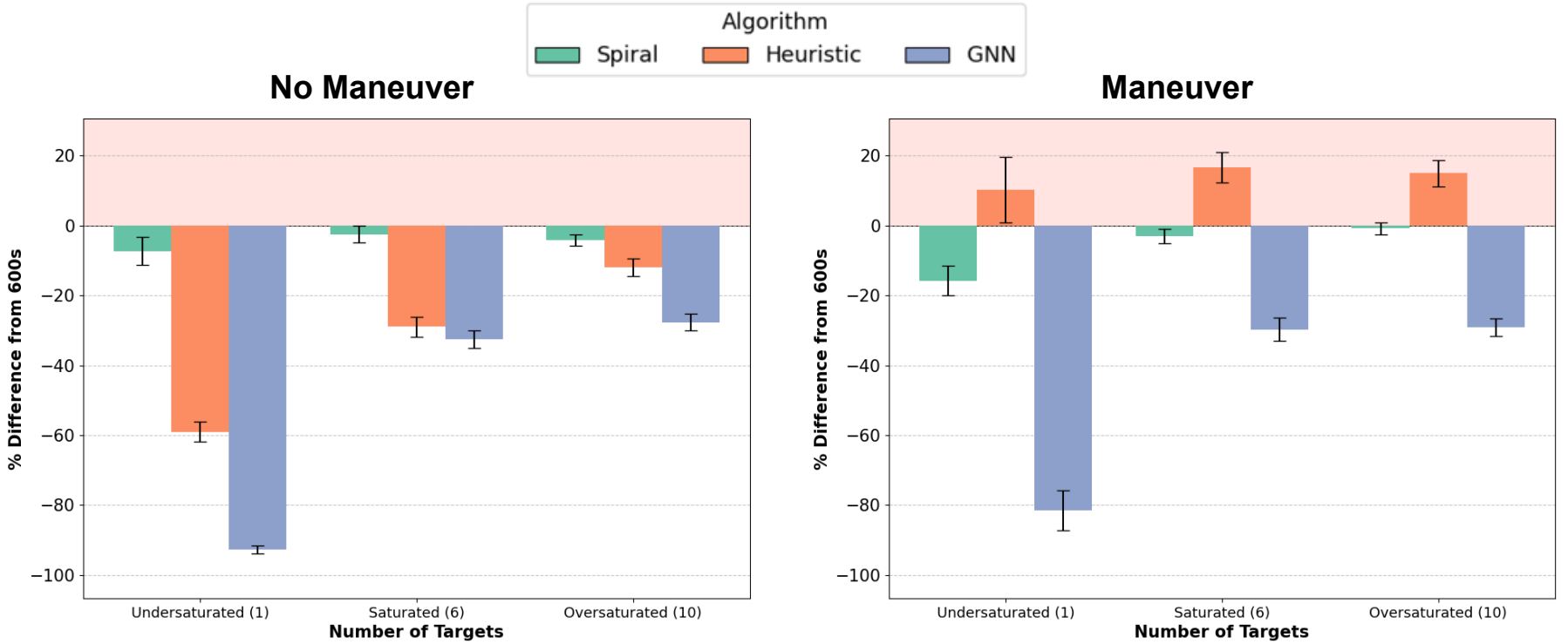
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Time to First Detection

- 500 Monte Carlo runs
- 30 min episodes



System Detection Rate

- 500 Monte Carlo runs • 30 min episodes

